

EV Charging and Grid Management Report

Introduction

Electric vehicle charging can substantially affect residential energy usage and peak demand.

Off-Peak Charging

Charging EVs during off-peak hours reduces stress on the electrical grid and lowers electricity costs.

Smart Charging

Smart charging systems dynamically adjust charging speed based on grid conditions and energy pricing.

Solar-Powered EV Charging

Integrating rooftop solar systems with EV chargers can significantly reduce grid dependency.

Future Smart Grids

Advanced smart grids use AI and predictive analytics to optimize energy distribution and EV charging behavior.

Metric	Example Value
Average Solar Efficiency	21%
Recommended AC Temperature	24°C
Peak Solar Hours	10 AM - 3 PM
Average EV Charging Duration	4-6 Hours